



EcoChef

AI-Powered Smart Kitchen Assistant

An Intelligent Recipe Generation System Using Computer Vision and Generative AI for Food Waste Reduction

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Project Overview

EcoChef addresses a common household problem: food waste caused by the inability to identify practical recipes from available ingredients with no existing tool that works offline or supports multilingual input.

The project's core objective is to develop a cross-platform AI kitchen assistant capable of detecting ingredients via camera or text input, suggesting matching recipes in real time, and operating fully offline. Four goals guided the development: **(G1)** accurate single-frame ingredient detection using a mobile-deployable YOLO model, **(G2)** recipe suggestions delivered within 2 seconds online and under 5 seconds offline, **(G3)** full Turkish-language support via a built-in ingredient dictionary and LRU-cached translation, and **(G4)** a production-ready, containerised system with secure authentication and persistent user data.

The system employs **YOLOv8** for real-time visual detection, **Llama GGUF** for on-device NLP inference, and the **Spoonacular** & **TheMealDB** APIs for access to 365,000+ recipes, backed by a **FastAPI** backend and **React Native** mobile client. Three research questions shaped the design: whether a single YOLOv8 model can reliably detect ingredients under varied conditions (**RQ1**), how effectively a quantised local LLM can substitute cloud-based NLP in resource-constrained environments (**RQ2**), and the impact of a multilingual ingredient dictionary on retrieval accuracy for nonEnglish users (**RQ3**).

Key results include **3 deployed AI pipelines**, **42 REST endpoints**, sub-second offline inference, an **80+ entry TR ↔ EN** ingredient dictionary, and full JWT security with PBKDF2-SHA256 (260k iterations).

Technology Stack

FastAPI
v0.135.2 · Python
3.12

React Native
v0.81 · Expo SDK
54

YOLOv8
Ultralytics
8.4.26

Llama GGUF
llama-cpp-python
0.3.18

MySQL 8.4
SQLAlchemy
2.0.49

JWT + PBKDF2
python-jose 3.5 ·
260k iter

deep-translator
v1.11.4 · LRU Cache
512

Expo Router
~v6.0 · File-based
navigation system

TypeScript
v5.9 · React 19.1

Docker Compose
3 container · Bridge
network

AI Pipelines

Pipeline 1 — Visual Analysis

Photo → YOLO v8 → Ingredient List → Spoonacular / TheMealDB → Recipe Suggestions

Pipeline 2 — Text Analysis

Turkish Text → TR→EN Translation → Parallel API → Recipe Suggestions

Pipeline 3 — Hybrid (Offline)

Visual + Text → YOLO v8 → Llama GGUF → Local Suggestions

Project Statistics

42

REST API Endpoints

365K+

Recipe Database

3

AI Pipelines

6

Database Tables

80+

TR→EN Ingredient Dictionary

2

Language Support (TR/EN)

Key Features

Image-based Ingredient Detection

Automatic ingredient detection from images is performed using YOLOv8, with a confidence threshold set to 0.40.

Bilingual Support & Full Localization

Supports English and Turkish languages with an 80+ ingredient dictionary, Google Translate LRU caching, and a fully implemented i18n localization system.

Weekly Meal Planning

7-day × 3-meal structured planning with AsyncStorage-based offline caching and week-based navigation support.

Favorites & Rating System

Recipe bookmarking and 1–5 star rating functionality with persistent storage in MySQL.

Offline Mode

Recipe suggestions are generated using a local Llama GGUF model without internet connectivity, with caching of the last 10 results for improved performance.

Diet & Nutrition Filters

Vegetarian, vegan, and gluten-free options with macronutrient-based filtering (calories, protein, carbohydrates, and fat).

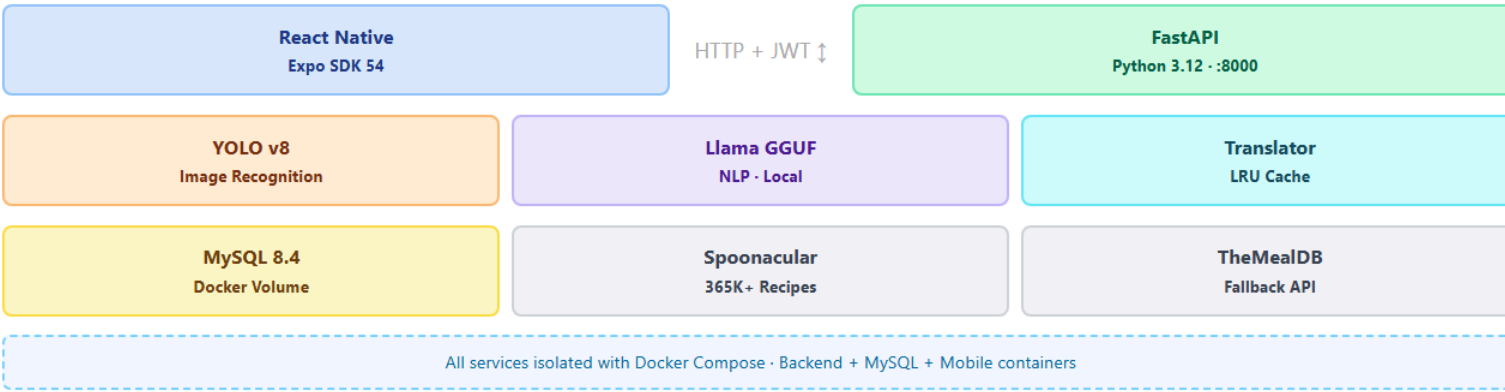
Shopping List

Automatically detects missing ingredients and adds them to the shopping list, with integration support for A101, Migros, Getir, and Trendyol platforms.

JWT Kimlik Doğrulama

PBKDF2-SHA256 password hashing (260K iterations), JWT-based authentication with 30-day token validity, token blacklist mechanism, and SecureStore-based secure credential storage.

System Architecture



Database Schema

USERS	FAVORITES	MEAL_PLANS
- PK id - username - password_hash - dietary_prefs	- PK id - FK user_id - recipe_id - recipe_name	- PK id - FK user_id - date · meal - recipe_name
RATINGS	SEARCH_HIST	REV_TOKENS
- PK id - FK user_id - recipe_id - score (1–5)	- PK id - FK user_id - query · mode - date	- PK id - token (UNIQUE) - revocation_date - JWT blacklist

EcoChef Screens

- Home Page** delivers AI-curated recipe suggestions
- Real Time Detection** uses YOLOv8 to identify ingredients through the device camera.
- The Explore screen** grants access to 365K+ recipes with advanced filters.
- Weekly Meal Planning** enables a full 7-day schedule with offline support
- Grocery Shopping** auto-generates shopping lists with direct retailer integrations.
- The Profile screen** handles user preferences, history, and security settings

Key API Endpoints

Method	Endpoint	Description
POST	/detect-ingredients/	Image → YOLO → Recipe (10/min)
POST	/analyze-text-ingredients/	Text → Spoonacular (20/min)
POST	/hybrid-suggest/	Visual+NLP (Offline)
GET	/recipes/random	Suggest random recipe
POST	/auth/login	Login → JWT token
GET	/favorites	Favorites list (JWT)
POST	/meal-plan	Add recipe to plan (JWT)
POST	/ratings	Rate recipe 1–5 (JWT)

Results & Contributions

- Multimodal AI Integration:** Combines image (YOLO) and text (Llama) models within a single unified platform.
- Fully Offline Hybrid Recommendation System:** Eliminates internet dependency by enabling complete offline functionality.
- Turkish NLP Support:** Low-latency translation using an 80+ entry ingredient dictionary and LRU caching mechanism.
- Production-Ready Architecture:** Implements Docker deployment, JWT-based security, rate limiting, and SQLAlchemy ORM.
- Comprehensive User Experience:** Includes features such as favorites, meal planning, shopping lists, and push notifications.